

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 01

Secure AWS Management Account

AWS Account Security | Root Identity Protection | MFA

AUTHOR	Michael Salazar
ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
RELEASE DATE	July 8, 2026
STATUS	Complete

Executive Summary

Created a dedicated AWS account for cloud security labs and immediately secured the root identity with a strong password and Multi-Factor Authentication (MFA). The configuration establishes a protected foundation for later identity, logging, monitoring, and governance projects.

Business Problem

A new cloud account begins with an identity that can control every resource and account-level setting. If that root identity is compromised, the entire AWS environment can be taken over. The first security objective was therefore to protect the root identity before deploying additional services.

Objectives

- Create a dedicated AWS account for isolated security lab work.
- Secure the AWS root identity with a strong password.
- Enable MFA before beginning day-to-day administration.
- Verify that no long-term root access keys exist.
- Establish a secure foundation for future AWS projects.

Environment

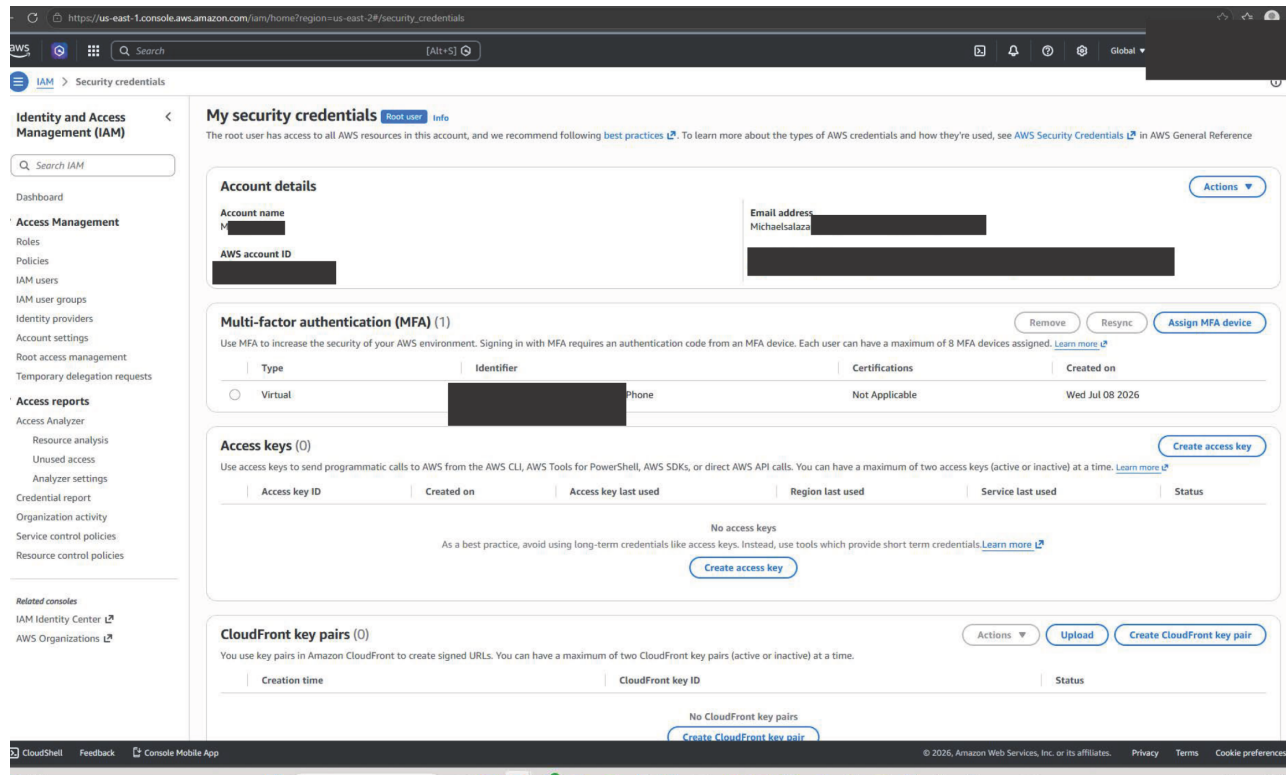
Cloud Provider	Amazon Web Services (AWS)
Environment	Personal cloud security lab
Identity	AWS root user
Authentication	Strong password + virtual MFA
Long-term Root Access Keys	None

Implementation Summary

1. Created a clean AWS account dedicated to security labs.
2. Configured a strong password for the root identity.
3. Enrolled a virtual MFA device and validated successful multi-factor sign-in.
4. Reviewed the root security credentials page and confirmed that no access keys were present.

Evidence

Figure 1 - Root Security Credentials and MFA



The redacted AWS Security Credentials page confirms one virtual MFA device is assigned to the root identity and shows zero root access keys. Sensitive account information was removed before publication.

Security Analysis

The AWS root identity is the highest-privilege identity in an account. Protecting it first reduces the risk that a single stolen password can lead to complete account compromise.

The evidence also shows that no root access keys were created. Avoiding long-term root credentials reduces the attack surface and supports the practice of reserving root access for rare account-level tasks.

Lessons Learned

A secure cloud environment begins before workloads are deployed. Account ownership, recovery paths, privileged authentication, and root credential handling are part of the security architecture.

Enabling MFA immediately established a secure baseline for every later AWS project in this portfolio.

Production Improvements

- Use a hardware security key or passkey for the root identity where practical.

- Store recovery details and root credentials in a controlled break-glass process.
- Configure billing and security contacts, budget alerts, and account-level monitoring.
- Use the root identity only for tasks that cannot be delegated.

Skills Demonstrated

AWS account provisioning	Root identity protection	Evidence sanitization
Security credentials review	Multi-factor authentication	Technical reporting
Access-key validation	Privileged access hygiene	Portfolio documentation

Resume Bullet

RESUME	Created and secured a dedicated AWS management account by protecting the root identity with MFA, avoiding long-term root access keys, and establishing a secure baseline for future cloud security projects.
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STAR Interview Story

Situation: I needed a clean AWS environment for hands-on cloud security work.

Task: Create the account without leaving the highest-privilege identity exposed.

Action: I created a dedicated lab account, configured a strong root password, enabled virtual MFA, and verified that no root access keys existed.

Result: The environment started from a protected baseline and was ready for delegated administration, logging, and later security controls.

Version History

v1.0.0	July 8, 2026	Initial standardized portfolio release. Created a dedicated AWS lab account, secured the root identity with a strong password and virtual MFA, verified that no root access keys were present, and documented the configuration with user-provided evidence.

References

Cloud Security Lab a Week (S.L.A.W.) - Creating Your AWS Lab Account.; User-provided, redacted AWS Security Credentials screenshot.